

PTC Uniform/Constant Analysis Coefficient

Scientific Rationale and Contract

SCI-PTC-COEFFICIENT-UNIFORM v0.1-draft.1 / r0.4

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Abstract

Revised draft for scientific-owner review. This shorter view explains the mathematical meaning, exact support, uncertainty limits, consumer boundaries, and choices still awaiting approval for a dimensionless uniform PTC analysis/gridding coefficient. Formal decision rules, failure scopes, and evidence mappings are carried by the paired engineering view. No implementation conformity, validation, readiness, Registry activation, policy, or default is claimed.

1 Program adherence and prior-work recovery

This implementation-blind revision follows only the released Stage A packet, the completed prior-work recovery synthesis, and the isolated scientific-owner review-1 supplement. It retains the frozen PTC coefficient taxonomy and lifecycle, the approved single-registry framework, the frozen MAP and JINC coefficient boundaries, the frozen NOI parent-binding limits, and the admitted VAL knowledge and named-use rules. Owner-requested recovery additionally cites only frozen SCI-PTC requirements 072, 090, 094, and 095 for atomic required outputs and ordinary-route rank provenance. No omitted parent material, implementation, test, audit, validation, or external scientific source was used.

The owner supports the fixed dimensionless-unity design, absence of a numerical parameter or default, separation from sample validity and uncertainty, separate MAP and JINC permissions, and the existing NOI limits in principle. The owner returned r0.2 for the targeted contract revisions rendered here and explicitly withheld complete contract approval and freeze. Every new family, profile, publication, permission, and boundary choice therefore remains an open proposal.

Revision r0.4 is a bounded consistency correction within scientific-owner round 1. It changes only the description of structural binding-record defects versus integrity of bytes in an already bound payload; it adds no source or scientific choice.

2 Scientific meaning of the uniform family

For every occurrence in its exact declared support, the PTC-owned coefficient is one. It is a transparent reference family: relative differences in a downstream contribution cannot originate from this coefficient. Other producer and consumer facts can still make contributions unequal or unusable. The constant is fixed at exact one because any positive common scale would be removed by mean-one normalization and would add a parameter with no surviving meaning.

Uniformity concerns only the positive PTC analysis/gridding coefficient. It does not assert equal signal quality, exposure, response, uncertainty, precision, significance, detector independence, or optimality. It does not restore an invalid sample, create support, or override a consumer gate. An occurrence authoritatively outside support has no coefficient under this family; absence is not numerical zero.

2.1 Handoff decisions and ownership

The proposed named-use profile asks whether one attempted handoff has the right selected family, exact parents and support, a faithful payload, and permission for one named consumer. The request to evaluate this profile is independent of the selected-family fact. Thus an explicit profile request with a known different selected family reaches UQ-01 and is ineligible; no explicit profile request creates no evaluation.

Payload fidelity is a producer-owned property shared by every handoff that uses the same immutable payload. It is composed from independently assessable scalar-value, unit, integrity, and support-membership facts. A decisive false in any one fact is not hidden by an unrelated unknown. Contradictory claims about the same fact remain a conflict. A faithful payload is not made corrupt by consumer policy: a complete authoritative denial makes only that named use ineligible, while missing or conflicting permission evidence leaves that use decision unavailable absent another decisive false.

The scope follows the object that is defective. A malformed reference in one handoff request affects only that attempted request. A defect in a shared authoritative support, scalar publication, family generation, or source binding affects every request that references that object. A named-consumer permission result affects that consumer and does not determine the other consumer.

Classification follows the established identity. A corrupt authoritative identity or binding record can prevent structural establishment. Once the payload publication identity and bindings are valid, corruption, digest failure, or decoder failure in its bound bytes is a payload-integrity fact under $Q_{\text{integrity}}$, never structural merely because the bytes are corrupt.

Request, applicability, eligibility, and realization answer different questions. A requested and applicable profile can finish with an ineligible or decision-unavailable result; that completed negative or unresolved result is still a successfully realized scientific record. Failure to finalize the record instead leaves no completed evaluation to reinterpret. No exception, override, or advisory fact can bypass identity, payload fidelity, or the named-consumer permission needed for that use.

2.2 Consumer boundaries

For MAP, the proposed family supplies only the existing PTC coefficient facet; MAP retains its own admission, placement, exposure, coordinate, and numerical classification rules. For JINC, the PTC factor becomes one while JINC retains the signed spatial factor, normalization, conditioning, and support. MAP and JINC permissions are independent and remain proposed rather than granted.

A stored one never supplies missing selection, authority, provenance, or permission. The explicit PTC-disabled route remains RTC-terminal and creates no PTC coefficient or MAP route. The ordinary-route positive-rank limits are stated from their admitted frozen source; this contract infers no identity, zero-rank, or other route family. NOI still requires the exact lossless rational parent for the selected MAP and coefficient generations and gains no new route or design rule.

3 Mathematical definition, support, and publication

3.1 Distinct identities and indices

Let a identify one application generation. Let q identify one exact, immutable, completed PTC transformed-product publication generation bound to a , its CAL parent, and one observation o . Let g identify one coefficient realization bound to q . The logical coefficient index i is an exact detector-time occurrence with stable identity ι_i . The symbol p is reserved for a consumer-owned spatial/map index; it never denotes a PTC product.

Each q declares exactly one publication-unit tag and bounds: a complete segment, a complete scan, or a bounded chunk wholly within one named segment/scan and observation. A unit never crosses observation, application, CAL-parent, segment/scan, or PTC-product-generation identity. The tagged unit size is explicit in q ; this contract supplies no hidden size default.

3.2 Exact support

The coefficient support and normalization population are

$$\mathcal{D}_{g,q} = \mathcal{S}_{g,q}^{\text{coef}} = \mathcal{S}_q^{\text{PTC,out}}. \quad (1)$$

They contain every and only exact occurrence in the completed output support of q . The population is finite, nonempty, and identity-unique. It may span all detectors, groups, arrays, networks, and times present in that publication unit; none silently defines a separate population. A singleton is valid. An empty population has no family instance. Observations and publication units are never pooled.

A required q is labeled complete only after all required components and links for that unit are durable, as required by frozen SCI-PTC v0.1/r0.5 requirement 072. This preserves the inherited atomic-output rule while allowing each completed segment, scan, or bounded chunk to be published and handed off independently. No coefficient handoff waits for unrelated publication units or for the whole observation to finish.

3.3 Uniform construction

For finite nonempty $\mathcal{D}_{g,q}$,

$$c_\star \equiv 1, \quad [c_\star] = 1, \quad (2)$$

$$\tilde{\omega}_i = c_\star, \quad i \in \mathcal{D}_{g,q}, \quad (3)$$

$$m_g = \frac{1}{|\mathcal{D}_{g,q}|} \sum_{j \in \mathcal{D}_{g,q}} \tilde{\omega}_j = 1, \quad (4)$$

$$\omega_i = \mathcal{N}_{\text{mean1}}(\tilde{\omega}_i; \mathcal{D}_{g,q}) \equiv \frac{\tilde{\omega}_i}{m_g} = 1. \quad (5)$$

The mean is a mathematical identity, not an instruction to execute a population-wide floating-point reduction of stored ones. A conforming realization may return exact unity analytically after the finite, nonempty support fact is established. Changing the normalization rule changes the family definition even though this family's result is one.

The family accepts no numerical parameter. A request carrying another constant, including another positive constant whose scale would cancel, is not an instance of this family and cannot be clipped, ignored, or normalized into acceptance.

3.4 Compact exact association

Let R_q be the immutable authoritative target-support reference in Equation 1. Its identity, source, generation, scope, and digest are structurally bound to q , and

$$\text{resolve}(R_q) = \{\iota_i : i \in \mathcal{D}_{g,q}\}. \quad (6)$$

A compact scalar publication carries a distinct stored support reference $\hat{R}_g$. Its membership fact is true only when the independently checked $\hat{R}_g$ exactly identifies R_q , including matching source, generation, scope, and digest. Equation 6 defines the target association; $\hat{R}_g$ is evidence that the payload realizes it without a copied member list. If a representation instead enumerates members, its complete stored manifest M_g must satisfy

$$M_g = \{(\iota_i, 1) : i \in \mathcal{D}_{g,q}\} \quad \text{with each identity exactly once.} \quad (7)$$

A missing or conflicting authoritative R_q prevents structural establishment. After R_q is valid, a wrong or mismatched stored $\hat{R}_g$, or an omission, addition, or repetition in M_g , is a payload-membership false. An unavailable stored reference or enumeration check is unknown; contradictory claims about that same stored fact are conflict. Row order, shape, cardinality, detector label, coordinate equality, or container position cannot replace exact identity.

Family identity, scalar value, unit, integrity, q -to- g compatibility, and named permission are immutable shared facts. They may be evaluated once for their declared scope. Each occurrence-level handoff remains logically associated with

those shared evaluations through exact references plus its own $i \in \text{resolve}(R_q)$ and request identity. Neither the coefficient array nor a separate full profile record must be materialized for every occurrence.

3.5 Consumer consequences

At the JINC boundary, p is JINC's spatial index and

$$w_{ip} = \kappa_{ip}\omega_i = \kappa_{ip}. \quad (8)$$

This equality transfers no ownership of κ_{ip} , its sign, or JINC support.

For the already selected NOI design on an exact MAP-admitted contribution population, p is the MAP spatial index, C_p is MAP's occurrence set for that spatial index, and

$$a_{pi} = G_{pi}\gamma_i = G_{pi}, \quad B_d = \sum_p \sum_{\substack{i \in C_p \\ \text{detector}(i)=d}} a_{pi}. \quad (9)$$

This does not define G_{pi} , change MAP arithmetic, or alter the NOI design law.

4 Scientific assumptions, uncertainty, and route limits

The family assumes only a completed, exact PTC publication generation q with identifiable finite nonempty output support. It assumes no statistical distribution, stationarity, independence, equal detector response, equal exposure, residual model, sensitivity model, or covariance model. The coefficient is defined rather than estimated and ranks no data quality.

The exact policy constant introduces no coefficient sampling uncertainty. This does not assign zero to signal, calibration, response, covariance, support-selection, family-selection, representation, or performance uncertainty. Unknown remains unknown, and absent covariance does not mean independence.

4.1 Ordinary-route and rank provenance

Frozen SCI-PTC v0.1/r0.5 requirement 090 defines the admitted first ordinary PTC route: the parent q is an explicitly requested calibrated- x , configured-rank, group-local PCA/SVD operation with exact CAL parent, segment, output identity, lifecycle, and failure state. Requirement 094 requires an integer rank at least one for each group, or one explicitly common positive rank applied to named groups; every group verifies realizability on its own support and degeneracy state. Requirement 095 says rank zero, noninteger rank, or an unrealizable positive rank fails closed with exact group and cause. It does not become centering-only output, get clipped or substituted, become PTC-disabled, select RTC-terminal export, or produce a map.

Consequently, this coefficient can be instantiated on the ordinary route only from a completed q produced after those rank obligations pass. The PTC-disabled route remains separately RTC-terminal and produces no PTC coefficient. These inherited statements provide no authority for an identity PTC route, a zero-rank route family, or any other route.

4.2 Family rules, realizations, and immutable change

A family-rule version defines the constant, unit, normalization operator, support rule, representation permissions, and profile semantics. Changing one of those scientific rules creates the appropriate versioned family, profile, permission, or boundary successor.

Applying unchanged rules to a different application generation, observation, publication unit, or exact support population creates new immutable q and g realizations under the same family version. It does not version the family. A published realization is never rewritten: a corrected or changed actual support produces a new realization and preserves the earlier record. Numerical equality alone does not authorize cross-generation borrowing.

5 Scientific choices still awaiting owner approval

Owner review round 1 supports the basic uniform design in principle and withholds approval of the complete contract. The owner still must adopt, revise, or reject:

- the registered exact-unity family and analytic mean-one convention;
- the exact publication-unit support rule, compact support reference, shared-fact association, and incremental handoff proposal;
- the named-profile identity, independent request rule, four fidelity facts, mixed-evidence composition, nonexceptionability, and failure scopes;
- the separate MAP and JINC grants and their r0.4 boundary successors;
- immutable realization, replay, supersession, uncertainty, and Registry source-binding rules; and
- the proposed actual-owner field, Grant Wilson acting for SCI-PTC.

The first five groups are complete author proposals rather than missing definitions. The final item remains precise question UFC-Q-001 and blocks active Registry/profile binding. No implementation conformity, observational performance, production readiness, Registry activation, default, or complete scientific freeze follows from this revision.

6 Consequences that distinguish the proposal

A known non-one scalar or authoritative corruption makes the shared payload invalid even when another fidelity fact is unknown. An unexplained unreadable payload remains unknown. A faithful payload paired with a MAP denial remains faithful while MAP use is ineligible; missing MAP permission evidence leaves the MAP decision unavailable. These distinctions preserve causes instead of turning every failure into either corruption or absence.

A compact exact support reference is scientifically sufficient when it binds the correct immutable support. Enumerated storage remains permitted but must contain every and only each target identity once. Shared facts can be reused through exact references, while a malformed occurrence request stays local. These rules preserve exact association without requiring an array of ones or a separately copied decision record for every sample.

The proposal predicts identical coefficient values across independently completed segment, scan, or chunk publications, while their realization identities remain distinct. It also predicts that a valid coefficient does not rescue invalid signal data, change MAP or JINC gates, or complete a missing NOI parent. These are future conformance tests, not evidence about an implementation.